The heat equation

Now that we have a basic understanding of boundary value problems and separation of variables for the product solution, we can start talking about the heat equation.

The **heat equation** is actually the partial differential equation we've been working with this whole time,

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

The heat equation models the flow of heat across a material, which is why we call it the heat equation. But many other real-world scenarios, unrelated to heat, also turn out to be modeled well by the heat equation.

Remember, we've already learned how to use the product solution and separation of variables to break the heat equation into two ordinary differential equations.

$$\frac{dw}{dt} = -k\lambda w(t)$$

$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

with $v(0) = 0$ and $v(L) = 0$

Solving the heat equation

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Even though the second order equation takes a little more work, we'll solve it first, because we'll need its solution when we solve the first order equation. If we rewrite the second order equation

$$v'' + \lambda v = 0$$

and its associated characteristic equation, we get

$$r^2 + \lambda = 0$$

$$r^2 = -\lambda$$

$$r = \pm\sqrt{-\lambda}$$

With the roots of the characteristic equation defined this way, we have to consider three possible scenarios, $\lambda = 0$, $\lambda < 0$, and $\lambda > 0$. In each case, the general solution of the second order equation is given below.

$$\lambda = 0 \qquad v(x) = c_1 + c_2 x$$

$$\lambda > 0 \qquad v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

$$\lambda < 0 \qquad v(x) = c_1 \cosh(\sqrt{-\lambda}x) + c_2 \sinh(\sqrt{-\lambda}x)$$

$$\text{(or } v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}\text{)}$$

For $\lambda = 0$ we'll substitute $v(0) = 0$ to get

$$0 = c_1 + c_2(0)$$

$$0 = c_1$$

Then substituting $c_1 = 0$ and $v(L) = 0$ gives

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$$0 = 0 + c_2 L$$

$$c_2 = 0$$

So we end up with the trivial solution $v(x) = 0$, and as a result $u(x, t) = 0$.

For $\lambda > 0$ we'll substitute $v(0) = 0$ to get

$$0 = c_1 \cos(\sqrt{\lambda}(0)) + c_2 \sin(\sqrt{\lambda}(0))$$

$$0 = c_1$$

Then substituting $c_1 = 0$ and $v(L) = 0$ gives

$$0 = 0 \cos(L\sqrt{\lambda}) + c_2 \sin(L\sqrt{\lambda})$$

$$0 = c_2 \sin(L\sqrt{\lambda})$$

Assuming that $c_2 \neq 0$ (otherwise we would still have the trivial solution $u(x, t) = 0$), we get

$$\sin(L\sqrt{\lambda}) = 0$$

$$L\sqrt{\lambda} = n\pi \qquad \text{with } n = 1, 2, 3, \ldots$$

$$\sqrt{\lambda} = \frac{n\pi}{L}$$

$$\lambda = \left(\frac{n\pi}{L}\right)^2$$

Which means that the general solution equation for the $\lambda > 0$ case becomes

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$$v(x) = (0)\cos\left(\sqrt{\left(\frac{n\pi}{L}\right)^2}\,x\right) + c_2 \sin\left(\sqrt{\left(\frac{n\pi}{L}\right)^2}\,x\right)$$

$$v(x) = c_2 \sin\left(\frac{n\pi x}{L}\right) \qquad \text{with } n = 1, 2, 3, \ldots$$

And for $\lambda < 0$ we'll substitute $v(0) = 0$ to get

$$0 = c_1 e^{-\sqrt{-\lambda}(0)} + c_2 e^{\sqrt{-\lambda}(0)}$$

$$0 = c_1 + c_2$$

Then substituting $c_1 = -c_2$ and $v(L) = 0$ gives

$$0 = -c_2 e^{-L\sqrt{-\lambda}} + c_2 e^{L\sqrt{-\lambda}}$$

$$0 = -c_2(e^{-L\sqrt{-\lambda}} - e^{L\sqrt{-\lambda}})$$

This equation tells us that either $c_2 = 0$ or that

$$0 = e^{-L\sqrt{-\lambda}} - e^{L\sqrt{-\lambda}}$$

$$e^{-L\sqrt{-\lambda}} = e^{L\sqrt{-\lambda}}$$

$$-L\sqrt{-\lambda} = L\sqrt{-\lambda}$$

$$2L\sqrt{-\lambda} = 0$$

But because this is the $\lambda < 0$ case, we know $L \neq 0$, and $\lambda \neq 0$ which means $e^{-L\sqrt{-\lambda}} - e^{L\sqrt{-\lambda}} \neq 0$. Therefore, the only way this equation is true is when $c_2 = 0$, and we end up with the trivial solution $u(x, t) = 0$.

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So the $\lambda > 0$ case is the only one giving a non-trivial solution, and we have just

$$\lambda = \left(\frac{n\pi}{L}\right)^2$$

$$v(x) = c_2 \sin\left(\frac{n\pi x}{L}\right) \qquad \text{with } n = 1, 2, 3, \ldots$$

The first order ordinary differential equation is a linear equation in standard form,

$$w' + k\lambda w = 0$$

so we know the solution is

$$w = Ce^{-k\lambda t}$$

$$w = Ce^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

Putting our results together from the first and second order equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u(x,t) = c_2 \sin\left(\frac{n\pi x}{L}\right) Ce^{-k\left(\frac{n\pi}{L}\right)^2 t} \qquad \text{with } n = 1, 2, 3, \ldots$$

While c_2 will be different for each n, the constant $C \times c_2$ will also depend on n, so let's rename it as B_n.

$$u_n(x,t) = B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t} \qquad \text{with } n = 1, 2, 3, \ldots$$

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We've written the equation for the solution with u_n and B_n to indicate that we'll get a different solution for each value of n.

The Fourier sine series

What we may be noticing by this point is that the solution to the heat equation is starting to look a lot like a Fourier sine series. In fact, if we now pair the general solution to the heat equation with a specific initial condition $u(x,0) = f(x)$, we'll be able to arrive at the Fourier sine series.

Remember that the general solution $u(x, t)$ is an equation defined for u in terms of the independent variables x and t, where we think about x as a spatial variable and t as a time variable.

So when we define an initial condition $u(x,0) = f(x)$, it's defining the function at time $t = 0$, but the spatial variable x is left to vary. In contrast, the boundary conditions $u(0,t) = 0$ and $u(L, t) = 0$ define the function at the spatial locations $x = 0$ and $x = L$, but leave time t to vary.

So let's imagine that we've been given the initial condition

$$u(x,0) = 2\sin\left(\frac{3\pi x}{L}\right)$$

Matching this up to the general solution of the heat equation, we can identify $n = 3$ and therefore $B_n = B_3 = 2$. So given this initial condition, the solution to the heat equation would be

$$u_3(x, t) = 2\sin\left(\frac{3\pi x}{L}\right)e^{-k\left(\frac{3\pi}{L}\right)^2 t}$$

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Now, we've talked before about the principle of superposition, which tells us that the sum of two solutions to a differential equation is itself also a solution. As an example, if we found a second solution, let's call it $u_5(x,t)$, in addition to this $u_3(x,t)$ solution, we could add the two solutions together, and we know that $u_3(x,t) + u_5(x,t)$ is also a solution to the heat equation.

For instance, imagine we've been given the initial condition

$$u(x,0) = 5\sin\left(\frac{2\pi x}{L}\right) - 3\sin\left(\frac{6\pi x}{L}\right)$$

Matching this up to the general solution of the heat equation, we can identify $n = 2$ and therefore $B_n = B_2 = 5$ for the first term, and $n = 6$ and therefore $B_n = B_6 = -3$ for the second term. Which means that, given this initial condition, the solution to the heat equation would be

$$u(x,t) = u_2(x,t) + u_6(x,t) = 5\sin\left(\frac{2\pi x}{L}\right)e^{-k\left(\frac{2\pi}{L}\right)^2 t} - 3\sin\left(\frac{6\pi x}{L}\right)e^{-k\left(\frac{6\pi}{L}\right)^2 t}$$

In fact, we can add as many solutions together as we'd like, and that sum will also be a solution. To cut right to the point, we can add an infinite number of solutions,

$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

and this will also be a solution to the heat equation. And if we apply any initial condition where $t = 0$, then the solution becomes

$$u(x,0) = f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

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But, all of a sudden, we realize that we have the Fourier sine series representation of the initial condition. In other words, as long as the initial condition $f(x)$ is piecewise smooth, we can find B_n using the formula from the Fourier series lessons,

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

and we'll get a general solution to the heat equation that also satisfies the initial condition $u(x,0) = f(x)$.

With all of this finally in place, let's look at an example so that we can see the complete process for solving the heat equation with a specific initial condition.

Example

Solve the boundary value problem, if $u(x,0) = 50$, $u(0,t) = 0$, and $u(L,t) = 0$.

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

We already know that this is the heat equation with a boundary condition given by $f(x) = 50$. Which means the coefficients B_n are

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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$$B_n = \frac{2}{L}\int_0^L 50 \sin\left(\frac{n\pi x}{L}\right)\, dx$$

$$B_n = -\frac{100}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

$$B_n = -\frac{100}{n\pi}\cos(n\pi) - \left(-\frac{100}{n\pi}\cos(0)\right)$$

$$B_n = -\frac{100}{n\pi}(-1)^n + \frac{100}{n\pi}$$

$$B_n = \frac{100(1-(-1)^n)}{n\pi}$$

$$B_n = \frac{100(1+(-1)^{n+1})}{n\pi}$$

and the solution to the boundary value problem is

$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = \frac{100}{\pi}\sum_{n=1}^{\infty}\frac{1+(-1)^{n+1}}{n}\sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

The Fourier series and Fourier cosine series

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Using a very similar process to the one we used earlier when we built the Fourier sine series solution to the heat equation, we can also build a Fourier cosine series solution,

$$u(x,t) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$A_0 = \frac{1}{L}\int_0^L f(x)\ dx$$

$$A_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

as well as a Fourier series solution that combines the sine and cosine series.

$$u(x,t) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t} + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k\left(\frac{n\pi}{L}\right)^2 t}$$

$$A_0 = \frac{1}{2L}\int_{-L}^L f(x)\ dx$$

$$A_n = \frac{1}{L}\int_{-L}^L f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$B_n = \frac{1}{L}\int_{-L}^L f(x)\sin\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

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